

Data Cleaning & EDA Report: Sales Dataset

Dataset: 183 raw orders, Jan–Jun 2025 → 180 after cleaning

Data Cleaning

- Duplicates removed: 3 (183 → 180 rows)
- Missing values: 9 blank `Units`, filled with 0
- Text standardized: Region, Salesperson, Product, Category (spacing/capitalization)
- Validation: no negative prices, no future dates, no unexpected categories

Core KPIs

Metric	Value
Total Sales	₹90,53,158
Total Units Sold	1,129
Total Orders	180
Average Order Value	₹50,295

Sales Breakdown

By Region: West ₹29.3L > North ₹26.8L > South ₹20.4L > East ₹14.0L **By Category:** Electronics ₹62.96L (70%) > Furniture ₹26.83L (30%) > Stationery ₹0.74L (<1%) **By Salesperson:** Ravi Kumar ₹17.28L (top) → Sneha Das ₹5.02L (bottom)

Outlier Detection

- Method: IQR (1.5x rule)
- 22 of 180 orders (12%) flagged as outliers
- All are large Laptop Pro 14 / 27in Monitor / Standing Desk orders — genuine large sales, not errors

Correlation

Pair	Correlation
Units vs UnitPrice	-0.26
UnitPrice vs Sales	0.73
Units vs Sales	0.13

→ Revenue is driven by **price**, not **quantity**

Trend Analysis

Month	Sales	MoM Change
Jan	₹13.72L	—
Feb	₹10.56L	-23.0%
Mar	₹19.17L	+81.5%
Apr	₹8.92L	-53.5%
May	₹25.46L	+185.3%
Jun	₹12.69L	-50.2%

- Linear trendline slope: +₹83,600/month (positive, but volatile — only 6 data points)

Basic Statistics

Statistic	Value
Mean	₹50,295
Median	₹11,996
Mode	₹0
Std Dev	₹95,215

By Region (mean / std dev):

Region	Mean	Std Dev
East	₹30,367	₹45,883
North	₹59,562	₹121,132
South	₹49,826	₹69,014
West	₹61,107	₹118,876

Key Insights

- Electronics = 70% of revenue
- Revenue driven by price, not volume
- Median (₹11,996) is more representative than mean (₹50,295) — mean skewed by outliers
- East region underperforms on both revenue and order volume

5. Jan-Jun trend is positive but too volatile (6 months, one strong month) to confirm as a reliable pattern